THARINDU KOTHALAWALA

Data Science Undergraduate | Machine Learning & AI Enthusiast | Mechatronics Engineer

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PROFESSIONAL SUMMARY

Data Science undergraduate at Sri Lanka Technology Campus (SLTC) with hands-on experience in machine learning, data engineering, and AI agent development, and an NVQ 5 background in Mechatronics, Robotics and Automation Engineering. Skilled in Python, Pandas, Scikit-learn, and TensorFlow, with practical project experience covering exploratory data analysis, predictive modeling, explainable AI, and big data pipelines. Completed the Machine Learning Specialization by DeepLearning.AI and Stanford University. Seeking a Data Science or Machine Learning role to apply strong analytical, programming, and problem-solving skills to real-world data-driven systems.

SKILLS

Programming Languages: Python, SQL, Java, JavaScript, TypeScript, Dart, C++

Machine Learning & Data Science: Machine Learning, Deep Learning, TensorFlow, Pandas, NumPy, Scikit-learn, XGBoost, SHAP, Feature Engineering, Model Evaluation, Regression, Classification, Exploratory Data Analysis (EDA), Statistical Analysis, Time-Series Analysis, Data Cleaning, Data Preprocessing

Artificial Intelligence: AI Agents, Large Language Models (LLMs), Prompt Engineering, LangGraph, OpenAI API, Decision Systems, Process Automation

Big Data & Databases: PySpark, MongoDB, PostgreSQL, SQLite, Firebase, Data Pipelines, RFM Analysis, Market Basket Analysis

Business Intelligence & Visualization: Power BI, Matplotlib, Jupyter Notebook

Backend Development & APIs: FastAPI, REST APIs, SQLAlchemy, Authentication, CRUD Operations

Tools & Platforms: Git, GitHub, GitHub Actions, Docker, DagsHub, VS Code, Streamlit

WORK EXPERIENCE

Student Intern, LAUGFS Corporation (Rubber) Ltd

Horana, Western Province, Sri Lanka | January 2023 - July 2023

- Gained hands-on industrial and technical exposure within a manufacturing environment as part of NVQ practical training.

Laboratory Assistant, Lanka Walltiles PLC (LANKATILES)

Colombo District, Western Province, Sri Lanka | January 2018 - June 2018

- Supported laboratory testing and quality-control processes on a part-time, on-site basis.

PROJECTS

Aladdin - AI Forex Trading Assistant (Individual Project, Ongoing)

Python, FastAPI, SQLAlchemy, AI Agents, Docker, GitHub Actions

- Built a multi-agent AI assistant for forex market analysis, risk management, trade planning, journaling, and explainable AI coaching.
- Designed a FastAPI and SQLAlchemy backend with CI/CD via GitHub Actions and containerized deployment using Docker.

SmartCare Hospital AI (Group Project, Ongoing)

Python, Pandas, Scikit-learn, XGBoost, SHAP, Streamlit

- Developing a machine learning pipeline to predict patient disease risk, including data preprocessing and model evaluation.
- Implementing explainable AI using SHAP and an interactive Streamlit application for real-time predictions.

Scalable Retail Analytics Pipeline (Group Project, Completed)

Python, PySpark, MongoDB, Big Data, Jupyter Notebook

- Built an end-to-end big data pipeline processing over 541,000 retail transactions using Bronze, Silver, and Gold data layers.
- Performed RFM analysis and market basket analysis, storing curated results in MongoDB for downstream analytics.

eBay Sales Business Intelligence (Individual Project, Completed)

Python, Pandas, Scikit-learn, Jupyter Notebook, Power BI

- Analyzed real eBay sales data through data cleaning, exploratory data analysis, and time-series analysis.
- Built machine learning models and an interactive Power BI dashboard to surface sales insights.

Lanka Smart Mart (Group Project, Completed)

Flutter, Dart, Firebase, SQLite, Google ML Kit

- Developed a full-stack e-commerce mobile application with Firebase services, offline SQLite storage, and authentication.
- Integrated AI-powered product recognition using Google ML Kit and built inventory management features.

QuizServer and Student Management App (Completed)

Electron, JavaScript, Java, Android, SQLite

- QuizServer: Electron desktop application for running quizzes over a local network, accessible via browser or QR code.
- Student Management App: Android application for managing student records with full create, read, update, and delete (CRUD) operations.

EDUCATION

BSc (Hons) in Data Science, Information Technology

Sri Lanka Technology Campus (SLTC) | October 2023 - February 2027 (Currently Studying)

NVQ 5 - Mechatronics, Robotics and Automation Engineering

Sri Lanka College of Technology, Maradana | April 2020 - July 2023 | Grade: Finalist

Diploma of Education - English Language and Literature

Aquinas College of Higher Studies | April 2018 - August 2018

Secondary Education

Bope Rajasinghe National School (BRNS) | January 2005 - December 2018

CERTIFICATIONS

- Machine Learning Specialization - DeepLearning.AI and Stanford University (July 2026)
- Advanced Learning Algorithms - DeepLearning.AI and Stanford University (July 2026)
- Unsupervised Learning, Recommenders, Reinforcement Learning - DeepLearning.AI and Stanford University (July 2026)
- Supervised Machine Learning: Regression and Classification - DeepLearning.AI and Stanford University (July 2026)

LEADERSHIP & ACTIVITIES

- Head Prefect, School Prefects' Board, Bope Rajasinghe National School: led a team of prefects, coordinating discipline, events, and student welfare initiatives.
- Member, Leo Club of Sri Lanka Technology Campus (August 2024 - Present): community service and volunteering initiatives.
- Member, AIESEC (March 2024 - Present): youth leadership and cross-cultural global volunteering programs.
- Member, LEO Club of Padukka (January 2022 - Present): local community service projects and volunteering.

REFERENCES

Dr. Chameera De Silva, PhD (UniMelb)

Head of the Department - Data Science / Lecturer
FOCIT - Faculty Industry Placement Coordinator
Member of Ethics Review Committee
Member of Technical Review Committee
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